Segments of well-orderings embed coherently: if two pairs of initial segments are isomorphic, the segment order on either side agrees

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the one-direction reduction, the transport through f, and the closing rigidity argument each sit verbatim in the span with their named internal citations.

Editorial note. The label lemordsegments names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Initial segments and ordinal membership. Same unit, \ollabel{wellordinitialsegment} (source line 799). At line 832: transporting the isomorphism of A_{a1} along f identifies it with the B-segment at f(a1).
- No well-order is isomorphic to its proper initial segment. Same unit, \ollabel{wellordnotinitial} (source line 776). At line 835: two isomorphic B-segments must sit over the same point, forcing b1 = f(a1).

Source reference. Original reference wellordinitialsegment: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 799–799. Prerequisite pointer invoked at proof line(s) 835; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference wellordnotinitial: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 776–776. Prerequisite pointer invoked at proof line(s) 838; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let* $\langle A, <\rangle$ *and* $\langle B, \lessdot\rangle$ *be well-orderings. If* $\langle A_{a_1}, <_{a_1}\rangle = \langle B_{b_1}, \lessdot_{b_1}\rangle$ *and* $\langle A_{a_2}, <_{a_2}\rangle = \langle B_{b_2}, \lessdot_{b_2}\rangle$*, then* $a_1 < a_2$ *iff* $b_1 \lessdot b_2$

Proof. We will prove *left to right*; the other direction is similar. Suppose both $\langle A_{a_1}, <_{a_1}\rangle = \langle B_{b_1}, \lessdot_{b_1}\rangle$ and $\langle A_{a_2}, <_{a_2}\rangle = \langle B_{b_2}, \lessdot_{b_2}\rangle$, with $f\colon A_{a_2} \to B_{b_2}$ our isomorphism. Let $a_1 < a_2$; then $\langle A_{a_1}, <_{a_1}\rangle = \langle B_{f(a_1)}, \lessdot_{f(a_1)}\rangle$ by wellordinitialsegment. So $\langle B_{b_1}, \lessdot_{b_1}\rangle = \langle B_{f(a_1)}, \lessdot_{f(a_1)}\rangle$, and so $b_1 = f(a_1)$ by wellordnotinitial. Now $b_1 \lessdot b_2$ as f's domain is B_{b_2}. □